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When online advertising backfires: how imperative call-to-action messages trigger psychological reactance

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Abstract

Purpose – Amid increasing advertising noise on social media, this study examines the phenomenon of reactance in response to call-to-action (CTA) imperative messages in social media advertisements and investigates whether users are more likely than not to avoid ads perceived as having a stronger persuasive intent. It examines the phenomenon of reactance in response to CTA imperative messages in social media advertisements.

Design/methodology/approach – A post-test control group experimental method was employed, where users were randomly exposed to manipulated and non-manipulated social media ads. The control group viewed ads without CTA imperative elements, while the experimental group encountered ads featuring multiple CTA messages. Both groups had identical segmentation attributes, with the same appearance and wording of ads except for the CTA elements. Statistical analysis was conducted on impressions and clicks.

Findings – Results show that ads lacking direct CTA imperative elements are more effective at eliciting clicks. Strongly persuasive CTA messages may provoke reactance, reducing ad effectiveness. These findings challenge common industry practices and highlight the importance of neutral, organic advertising strategies in social media environments.

Originality/value – This study enhances the understanding of the relationship between CTA imperative messages and user reactance. Extending Brehm's Psychological Reactance Theory (PRT) introduces the inductive reactance model (IRM), offering a deeper theoretical framework. The findings may lead marketers to reassess the use of imperative CTA messages, questioning practices that prioritize direct CTA.

Keywords Call-to-action imperative messages, Psychological reactance theory, Advertising reactance, Facebook a/b split test, Social media advertising, Experimental research, Field study

Paper type Research paper

1. Introduction

A social media user sits in front of the screen, scrolling through an endless stream of content. Hidden among organic posts, advertisers continuously target the user with carefully crafted messages—each designed to persuade, like precisely aimed bullets fired from an invisible weapon. Most of these persuasive shots go unnoticed, subtly influencing decisions beneath the surface of consciousness. However, as social media has evolved, so has the user. Having spent years navigating digital platforms, encountering countless advertisements in varying formulations, the user has become more aware of these tactics. As the online space becomes increasingly saturated with ads, advertisers respond by intensifying their efforts—using stronger, more imperative call-to-action (CTA) messages. "Buy now!", "Don't miss out!". These once-effective persuasive triggers are becoming more obvious, more forceful. But what happens when the user starts to recognize the pattern? Rather than passively accepting the message, heightened awareness may alter their response. A message perceived as too controlling might not drive engagement as intended but instead trigger psychological reactions that disrupt persuasion. This phenomenon raises a critical question, introducing a novel perspective:

Do imperative CTA messages enhance ad efficiency, or do they provoke reactance and undermine their intended persuasive goal?

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The digital advertising landscape has undergone a fundamental transformation with the rapid growth of social media (Wang, 2021). Users now spend hours daily on these platforms, making them prime targets for advertisers (Ji *et al.*, 2022). The financial investment in online advertising reflects this shift: global ad spending has surged from $506.43 billion in 2021 to $667.58 billion by 2024 (eMarketer, 2023), while social media user numbers have expanded from 3.1 billion to 5.17 billion in the same period (Meltwater, 2024). Given this exponential growth, social media—especially Facebook—has become one of the most critical arenas for digital marketing (Setiawan *et al.*, 2024; Frankwatching, 2023).

However, with increasing ad saturation on social media, consumers are constantly exposed to persuasive messaging, particularly CTA imperatives that prompt immediate responses (Oltra *et al.*, 2022; Chen *et al.*, 2020). These messages—such as "Buy now!" or "Sign up today!"—are designed to increase ad efficiency and drive conversions. Prior research has extensively examined their effectiveness (e.g. Braca and Dondio, 2023; Vafainia *et al.*, 2019), emphasizing their positive influence on consumer behavior. However, the effectiveness of CTA imperatives may not be as straightforward as previous research suggests.

This literature gap is particularly relevant in light of increasing ad saturation and growing user awareness. While prior research has shown positive effects of CTA imperatives on user behavior (Vafainia *et al.*, 2019) and attitudes (Oltra *et al.*, 2022), little is known about their potential to trigger psychological resistance when perceived as overly controlling. In some cases, these messages may function like a backfired shot – intended to hit the target but instead triggering unintended reactance. When consumers perceive such imperatives as coercive, they may respond with psychological reactance—a defensive reaction to perceived threats to their autonomy (Brehm, 1966). Instead of enhancing ad efficiency, imperative CTA messages may deter users by creating a perceived threat to freedom, which can lead to reactance (Bilfinger *et al.*, 2023; Choi and Kim, 2022; Hu and Wise, 2021; Youn and Kim, 2019). This contradicts the widely held belief that direct commands always drive action. Existing models have not empirically assessed how reactance processes mediate the relationship between CTA language and advertising effectiveness in digital environments.

Our major contribution lies in addressing this gap by introducing the Inductive Reactance Model (IRM), an extension of Psychological Reactance Theory (PRT), inspired by Maxwell's (1873) concept of inductive reactance in physics. The IRM conceptualizes imperative CTAs as high-voltage inputs that generate resistance in the persuasive process. The novelty of this study stems from empirically measuring the effect of CTA imperatives on advertising reactance. By integrating PRT with a physics-based analogy, the IRM offers a novel lens for understanding user resistance in digital persuasion and provides fresh insight into the unintended consequences of directive messaging.

2. Conceptual background

2.1 Inductive reactance model (IRM)

As the user perceives more persuasive intent in advertisements, a mental shield begins to form, activated by imperative CTAs. The effectiveness of imperative CTA messages in social media advertising is influenced by psychological reactance, a phenomenon where individuals resist marketing efforts perceived as limiting their freedom of choice (Brehm, 1966; Godfrey *et al.*, 2011). To better understand this dynamic issue, we developed the *Inductive Reactance Model* (IRM) (see Figure 1), inspired by Maxwell's (1873) capacitive and inductive reactance electricity concepts from physics. The IRM model suggests that the mental inductor is activated by the overly intrusive CTAs, analogous to high voltage in an electrical circuit, generating resistance (reactance), diminishing the persuasive impact, and potentially delaying or nullifying the desired response.

In this framework, a company's advertising communication can be likened to alternating current (AC), with periodic changes in direction and intensity. The success of persuasion depends on whether the audience reacts like a capacitor or an inductor. In the first case, the

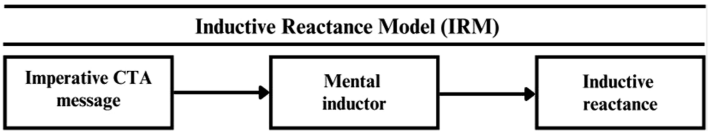


Figure 1. Inductive reactance model (IRM). Source: Authors' own work

audience reacts as we expect, "accepting" the current and going along with what we want (in this case, following the imperative). In the case of the inductor, the high intensity (imperative CTAs) creates resistance, slowing down or blocking the persuasive current (Maxwell, 1873). In communication terms, this resistance arises from perceived violations of autonomy, leading to cognitive opposition and reduced receptiveness to the message.

Social media platforms—especially Facebook—are oversaturated with advertisements, inherently limiting users' autonomy in choosing the content they engage with (Wang, 2024; Sheng *et al.*, 2023). Users' sense of free choice is an illusion (Baudrillard, 1981), which is further disrupted when imperative CTAs impose a directive force, increasing reactance. IRM suggests that reducing imperative CTA intensity can mitigate reactance, ultimately enhancing advertisement effectiveness. By challenging traditional marketing assumptions that favor strong CTAs, this perspective introduces a real novelty to the study.

The application of IRM provides a theoretical lens to examine how CTA imperative messages may lead users to resist advertisements on platforms like Facebook. This approach is particularly relevant in the evolving landscape of interactive marketing, where theoretical advancements are necessary to keep pace with industry changes (Wang, 2022).

2.2 Loading the gun: CTA imperatives

Just as a marksman carefully aims each shot, advertisers craft CTA messages to precisely hit their target as the person scrolls through social media. These messages act as the "bullet" of persuasion, where the strength and direction of the CTA determine whether it prompts engagement or flies off into reactance. CTA messages in advertising are visual elements or messages strategically designed to prompt desired action (Oltra *et al.*, 2022; Chen *et al.*, 2020). CTA messages primarily focus on directly shaping purchasing patterns. This can be achieved by presenting an appealing offer (call) that encourages customers to participate (take action) (Chae, 2021; Oltra *et al.*, 2022). These messages typically frame purchasing as urgent and necessary, leveraging psychological motivators such as scarcity and time pressure to drive engagement (Braca and Dondio, 2023; Belanche *et al.*, 2017; Kalman *et al.*, 2021). While prior research highlights their effectiveness in boosting engagement and conversions (Vafainia *et al.*, 2019; Borges-Tiago *et al.*, 2019), fewer studies have examined their potential adverse effects, particularly in contexts where users perceive highly directive messages as a threat to their freedom (Bilfinger *et al.*, 2023; Choi and Kim, 2022).

According to Vafainia *et al.* (2019), CTA messages are crucial incentives in corporate communication, aiding customers in brand-related information searches, reducing pre-purchase uncertainty, and enhancing purchase intentions. Fitzsimons and Lehmann (2004) similarly emphasize that minimizing decision-making effort boosts user confidence. Chen *et al.* (2020) analyzed Facebook native ads and identified common CTA forms like *"Learn more!"* and *"Buy now!"*, paralleling those found by Pennock-Speck and Fuster-Marquez (2014), such as *"Get this!"* and *"Taste the other!"*. Despite differences in wording, these imperatives serve the same function—to prompt action.

Imperatives express directive force, guiding users toward desired actions (Jary and Kissine, 2014; Miller *et al.*, 2007). Numerous studies have explored their use in advertising (Nisa *et al.*, 2022; Zjakic *et al.*, 2017; Labrador *et al.*, 2014). Pennock-Speck and Fuster-Marquez (2014) analyzed imperative messages in television advertisements, developing a five-category

classification based on their discourse functions. Imperative messages can grab attention (e.g. "Don't scroll down!"), encourage contact ("Send us a message!"), prompt acquisition ("Buy now!"), invite experience ("Try it out!"), or trigger other actions outside these categories.

This research examines CTA elements through a text-based approach, as Facebook's fixed ad format (e.g. button position, design, link display) limits distinctions in action encouragement. Text-based differentiation offers the most practical utility. Imperative messages were formulated based on Pennock-Speck and Fuster-Marquez's (2014) categorization theory to grab attention, encourage contact, prompt acquisition, invite experience, or trigger other unique actions.

2.3 Growing shield: Psychological Reactance Theory

As social media users become more aware of advertising tactics, they recognize when a message is designed to influence their decisions. When CTAs explicitly signal persuasion, users may perceive them as attempts to control their choices, prompting reactance instead of engagement. A persuasive message aims to strike its target and prompt the desired response, but if too forceful, it can backfire, triggering reactance instead of compliance. This psychological resistance acts as a shield, causing persuasive attempts to ricochet and backfire toward the sender, like a bullet bouncing off armor. This resistance is not just a behavioral response but follows a predictable psychological mechanism. As a result, the effectiveness of CTA imperative messages may be diminished by psychological reactance, where consumers resist marketing efforts they perceive as manipulative or limiting their freedom of choice (Brehm, 1966; Godfrey *et al.*, 2011). Instead of increasing efficiency, CTA messages may reduce the likelihood of achieving the desired effect or even cause negative reactions (Kalny and Walter, 2024; Reynolds-Tylus, 2019; Fitzsimons and Lehmann, 2004). Numerous studies have documented the harmful effects of intrusive marketing communications (Choi and Kim, 2022; Youn and Kim, 2019; Van Doorn and Hoekstra, 2013; Fitzsimons and Lehmann, 2004).

To address these challenges, the IRM offers a conceptual framework for understanding how imperative CTA messages can lead to psychological reactance. The IRM is the first model to examine how reactance operates specifically in response to CTA imperative messages in social media advertising, addressing the growing connection between reactance theory and social media, where personalized content created using users' data increasingly fosters mistrust and resistance (Lambillott and Poncin, 2022; Youn and Kim, 2019).

Although personalized ads are designed to be relevant, they can still provoke avoidance (Ham, 2017). Factors contributing to ad avoidance include skepticism, irrelevance, prior negative experiences, and perceived intrusiveness (Snyder *et al.*, 2024; Lee *et al.*, 2016). Most previous studies have only indirectly addressed the relationship between reactance and imperative messages in social media advertising, focusing instead on measuring factors (e.g. ad avoidance, intrusiveness, the threat to freedom) that are related to reactance (Bilfinger *et al.*, 2023; Hu and Wise, 2021). In the last few decades, only a few studies have investigated the relationship between reactance and imperative messages in some aspect (Kalny and Walter, 2024; Xu, 2014; Miller *et al.*, 2007). However, the key research gap is the absence of studies that specifically examine how CTA imperative messages influence user reactance. Addressing this gap makes the findings of this study particularly valuable for the field. To provide a systematic overview of relevant studies and their findings in the field, a table of literature has been created (see Table 1).

Reactance is commonly triggered by imperative language, which increases perceived threats to autonomy (Miller *et al.*, 2007). Promotional messages often contain controlling expressions such as "must," "should," "have to," and "necessary," as opposed to autonomy-supportive phrases like "maybe," "might," or "could" (Bilfinger *et al.*, 2023). Youn and Kim (2019) highlight the role of perceived intrusiveness, distinguishing two key subcomponents of reactance: negative cognitions and anger. Other research also links intrusive advertising to negative emotional and behavioral reactions, which are closely related to the emergence of

Table 1. Table of literatures in reactance and call to action imperatives

Topic	Study	Context	Method and sample size	Key findings
Reactance	Snyder *et al.* (2024)	Ad customization, voice assistants and reactance	Experiment, $n = 405$	Control afforded by customization reduces reactance; findings inform strategies to maximize customization's positive effects
Reactance	Bilfinger *et al.* (2023)	Freedom threat and reactance in climate change	Experiment, $n = 623$	Individual conditions increase counterarguing and reduce support for mitigation; policy-focused conditions lessen reactance
Reactance	Lambillott and Poncin (2022)	Online personalized advertising and reactance	Interviews, $n = 20$	Identifies three paradoxes of personalization; users navigate them through resolution and acceptance strategies
Reactance	Choi and Kim (2022)	In-stream video ads: perceived control	Experiment, $n = 217$	Ad skip options and time display improve control perception, reducing intrusiveness and irritation, enhancing attitudes toward ads
Reactance	Hu and Wise (2021)	Playable ads and perceived control	Experiment, $n = 199$	Increased perceived control alleviates freedom threat, reducing psychological reactance
Reactance	Youn and Kim (2019)	Reactance-related factors in Facebook ads	Survey, $n = 249$	Reduced autonomy increases ad intrusiveness; freedom threat positively impacts reactance, leading to ad avoidance
Reactance	Ham (2017)	Online behavioral advertising and ad avoidance	Survey, $n = 442$	Persuasion knowledge indirectly influences ad avoidance; privacy concerns mediate reactance; reactance and perceived personalization linked to avoidance
Reactance and Language	Kalny and Walter (2024)	Reactance and epistemically certain language	Experiment, $n = 676$	Imperative language increased anger and freedom threat; epistemically uncertain language reduced anger and reactance
Reactance and Language	Xu (2014)	High sensation value and reactance	Experiment, $n = 230$	High MSV enhanced ad effectiveness; controlling language contributed to anger; low controlling language improved message reception
Reactance and Language	Miller *et al.* (2007)	Reactance and Promotional Health Messages	Experiment, $n = 396$	Negative outcomes linked to controlling language; restorative postscripts improved outcomes; concrete language captured attention

(*continued*)

Table 1. Continued

Topic	Study	Context	Method and sample size	Key findings
Imperative Language	Nisa *et al.* (2022)	Food waste messaging (partly: imperatives)	Experiment, $n = 2,453$	More imperative messages (e.g. "Stop Food Waste") can elicit reactance and be counterproductive compared to milder persuasive approaches
Imperative Language	Zjakic *et al.* (2017)	Use of imperatives in web advertising	Multi-modal analysis, $n = 120$	Imperatives capture attention and generate interest without threatening politeness
Imperative Language	Labrador *et al.* (2014)	Persuasive language in online advertisements	Corpus-based analysis $n = 200$	High use of imperatives and second-person pronouns in online ads creates a sense of closeness with consumers
Imperative Language	Ren (2011)	Imperative clauses in English netvertising	Corpus-based analysis $n = 124$	Imperative clauses enhance cohesion in ads via structural and non-structural elements
Call to Action	Oltra *et al.* (2022)	Customer inspiration, call to action, and visual content	Experiment, $n = 2,626$	Brand followers consistently inspired; non-followers benefit more from posts without explicit brand mentions; carousel CTAs enhance inspiration
Call to Action	Freeman and Dardis (2022)	Disclosure modality, CTA, and customers' reactions	Experiment, $n = 482$	CTA messages enhance recognition, activate persuasion knowledge, and mitigate negative evaluations of ads
Call to Action	Vafainia *et al.* (2019)	Call to action and customers' purchase behavior	Panel analysis, $n = 179,525$	CTA DMs with incentives positively impact purchase probability; customer heterogeneity plays a significant role
Call to Action	Chen *et al.* (2020)	Call to action and consumers' recognition	Experiment, $n = 566$	Video format ads with "shop now" CTA enhance recognition; varying recognition levels correlate with differences in entertainment and sharing intent

Source(s): Authors' own work

reactance (Choi and Kim, 2022; Citalada *et al.*, 2022; McCoy *et al.*, 2016; Ying *et al.*, 2009). Their findings suggest that a decrease in perceived autonomy increases the perceived level of advertising intrusiveness. Consequently, both perceived advertising intrusiveness and the threat to freedom experienced while using Facebook positively impact reactance, which in turn affects ad avoidance. Reactance and ad avoidance are distinct but related: the former is a psychological response to perceived threats, while the latter reflects behavioral efforts to reduce ad exposure (Speck and Elliott, 1997).

Reactance is typically induced by persuasive language choices (Rains, 2013; Quick *et al.*, 2013). Dominant, controlling language, or perceived message vulnerability, can heighten perceived threats to autonomy (Kalny and Walter, 2024; Dillard and Shen, 2005; Li *et al.*, 2002). Kalny and Walter (2024) suggest that imperative language increases reactance, whereas epistemically uncertain language (e.g. "there is no other reasonable conclusion") can reduce perceived threats and enhance persuasive effectiveness (Quick and Considine, 2008).

With the growing prevalence of advertisements on Facebook, user perceptions of these ads have evolved (Wiese *et al.*, 2020). Native ads embedded in the Facebook news feed interrupt

users' use of Facebook and are therefore considered disruptive, as they hinder users' goals related to platform use (Ham *et al.*, 2021; Huang, 2019). This disruptive nature induces negative feelings, such as advertising irritation, which increases reactance and cognitive and behavioral ad avoidance in the interactive advertising environment (Li *et al.*, 2002; McCoy *et al.*, 2016). Based on this, we hypothesize that Facebook ads lacking CTA imperative messages may be more effective, as they restrict users' perceived freedom less and are less likely to trigger negative cognitions and behavioral responses that lead to reactance and ultimately ad avoidance.

H1. Users are less likely to click on social media advertisements that predominantly use imperative call-to-action messages.

3. Methodology

After developing the theoretical framework, Facebook A/B split testing was employed as a real-world online experimental methodology. Facebook a/b split tests have high ecological validity at both the independent and dependent variable levels, making them an appropriate methodology for testing the research question. To ensure rigor, a robust statistical method and precise experimental design were selected, adhering to strict scientific methodology for unbiased analysis, transparent reporting, and reproducibility. The study investigated the phenomenon of reactance induced by imperative CTA messages in social media advertisements, using clicks on tested ads as the measure. The study's manipulation test uniquely incorporated a LIWC analysis, categorizing and comparing the content of the tested texts across different dimensions.

3.1 Performance metrics

To test H1, the click metric was selected, as it is commonly used as a success measure in online advertising (see Table 2.). The number of clicks represents how often users clicked on elements of the advertisement that can redirect them to a website or a desired landing page. The click-through rate is typically used to evaluate the attractiveness of an advertisement (Järvinen and Karjaluoto, 2015).

Research indicates that the click metric most accurately measures the extent of user engagement with advertisements. Clicking on the ad will precede the fulfillment of all subsequent outcome metrics, making it the metric with the fewest distorting conditions compared to all other performance metrics.

3.2 Control/experimental group creation

A standardized algorithmic approach was employed using Facebook A/B split testing, creating control advertisements and experimental advertisements, which were developed based on information related to the brand profile of the advertiser stored in their Meta Business account. The advertisements were launched by a Hungarian jewelry retailer and targeted women aged 18–60 residing in Hungary.

The study utilized a real-world post-test control group experimental design, where participants were randomly assigned to either the control or experimental group. The

Table 2. Main metric

Main metric	Purpose
Number of clicks	*Click*: The number of times a user has clicked the ad

Source(s): Authors' own work

segmentation characteristics for users in both groups were identical, ensuring consistency in targeting criteria. The Facebook Business ad manager's a/b testing feature was instrumental in restricting each user to viewing only one version of the advertisement, either the control or the experimental, thereby preventing overlap. Participants were randomly selected from the brand's target audience, with the audience characteristics specifically defined by the brand. The experimental subjects were assigned to advertisements by Facebook's system based on pre-defined parameters, such as interests related to jewelry and online shopping. Facebook maintains interest profiles of users based on their online behavior, enabling advertisers to select these interests as targeting criteria in the ad manager. This functionality effectively ensured that the targeted women's interests aligned with the advertiser's business profile. These segmentation characteristics were consistent for both the control and experimental groups. Advertisements reached only relevant users. Each participant was assigned to a single group, ensuring no exposure to the other ad version. The control group advertisements did not contain any CTA imperative messages; they aimed to provide neutral messages. In contrast, the experimental group users encountered numerous CTA imperative messages. The appearance and phrasing of the advertisements were consistent in both cases, being identical except for the CTA imperative messages.

3.3 Ad design

In designing the advertisements, consistency was prioritized to minimize distortion in users' perceptions when viewing the ads. Both the control and experimental advertisements followed the same schema. Each Facebook ad included an image, associated text, and a CTA button, as depicted in Figure 2. While ad appearance varied slightly due to resolution and positioning, the message and creative content remained consistent.

The central element of the advertisement is the ad image [C], which dominates the visual appearance of the ad. However, additional information can be conveyed through the primary text [B] and the title and description fields [E]. The advertisement always began with the name of the Facebook page [A]—and this was also the brand name. The ad included the website URL [D] and a CTA button [F], which allowed users to navigate to the website. In designing the CTA imperative messages (Table 3.), a categorization of five types introduced by Pennock-

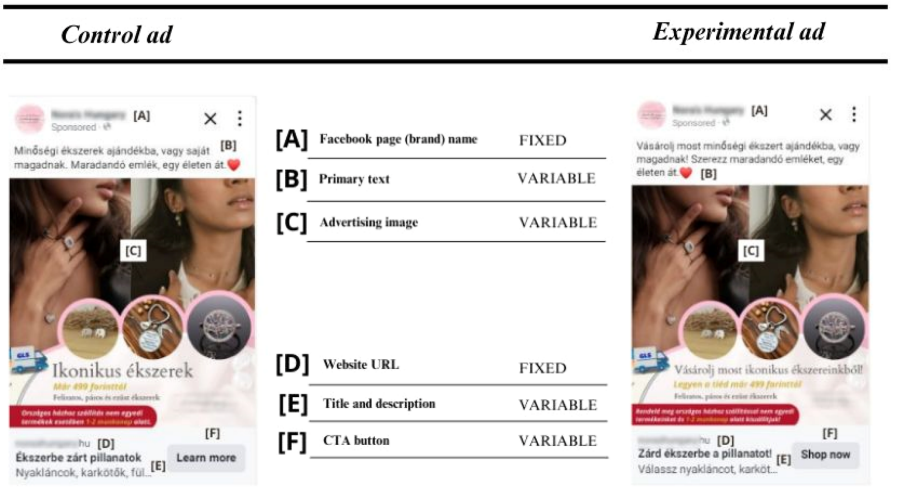


Figure 2. Example of control and experimental ad. Source: Authors' own work

Table 3. The elements of control and experimental ads

Elements of ads	Control Ad	Experimental Ad
[A]Facebook Page name	Brand name	Brand name
[B]Primary text	e.g - Quality jewelry as a gift or for yourself - A memory to last a lifetime. etc	e.g - Buy lasting jewelry now for yourself or as a gift! - Get a lasting memory for a lifetime! etc
[C]Ad creative image	e.g - Iconic jewelry - From HUF 499 - Inscribed, paired and silver jewelery - Nationwide home delivery for non-custom products within 1–2 working days - Made of premium quality materials etc	e.g - Shop now from our iconic jewelry! - Be yours from HUF 499! - Inscribed, paired and silver jewelry! - Order our non-custom products with nationwide home delivery and we will deliver them in 1–2 working days! etc
[D]Link of redirecting	Website URL	Website URL
[E]Title and description	e.g - Jeweled moments - Necklaces, bracelets, earrings, pendants, rings and more - Happy moments locked in jewelry	e.g - Embrace the moment! - Attention, products available for a limited time! - Choose now, while supplies last, necklaces, bracelets, earrings, rings and much more!
[F]CTA button	Learn more	Shop now!

Source(s): Authors' own work

Speck and Fuster-Marquez (2014) was followed, which serves to define imperative messages (Table 4.).

The study demonstrates that CTA messages can be attention-grabbing (e.g. "Don't scroll down!"), encourage engagement (e.g. "Send us a message!"), prompt acquisition of a product/service (e.g. "Buy now!"), invite experience (e.g. "Try it out!"), and call for other actions that do not fit into the above categories. Beyond the categorization by Pennock-Speck and Fuster-Marquez (2014), in crafting the messages for the experimental advertisements, CTA imperative messages were formulated to leverage the rhetorical power of urgency, repetition, and direct calls to action. As previously discussed, urgency, repetition, and creating a sense of scarcity are strong psychological motivators and are often used in imperative CTA messages (Kalman *et al.*, 2021; Oltra *et al.*, 2022; Braca and Dondio, 2023). CTA imperative messages without elements of urgency or repetition provided direct instructions for the desired action (e.g. "Make it yours!"). In the control advertisements, CTA imperative messages were omitted, presenting only neutral information and factual statements (Table 5.).

3.4 Manipulation check: text analysis of control and experimental ads (LIWC)

To test the effectiveness of the experimental manipulation, a linguistic analysis was performed using the LIWC dictionary (Tausczik and Pennebaker, 2010). The analysis compared control and experimental advertisements, demonstrating that the latter contained stronger persuasive elements.

The results (Table 6.) show that experimental advertisements exhibited higher values across all LIWC dimensions where data were available. Significant differences emerged in Social Words and Allure. The increased use of social expressions (e.g. "Surprise your loved ones") aligns with previous research indicating that social words enhance persuasion (Fenko

Table 4. Categorization of imperative messages by Pennock-Speck and Fuster-Marquez (2014) with examples

Imperative category	*Example*
Attention focus	Stop, don't keep scrolling!
Encourage contact	For unique jewelry, write us a message!
Encourage acquisition	Shop now from our iconic jewelry!
Encourage experience	Get a lasting memory for a lifetime!
Others	Hold on, the sale is about to start!

Source(s): Authors' own work

Table 5. Communication elements of control and experimental ads with examples

	Communication elements	Example
Control ads	Neutral, informing messages	In case of an order, we offer 2 days home delivery. Iconic jewelry
Experimental ads with imperatives	Urgency	Available only while supplies last!
	Repetition	Order, Buy, Acquire!
	Straightforward invitation for desired action	Order now with nationwide delivery!

Source(s): Authors' own work

Table 6. Result of LIWC analysis

Traditional LIWC dimension	Average for commercial language	Control ads text	Experimental ads text	Relative difference Calculated as (control-experimental)/ experimental
I-words (I, me, my)	*4.77*	0.00	0.00	0%
Positive tone	*3.96*	1.16	2.20	−47.27%
Negative tone	*1.10*	0.00	0.00	0%
Social words	*6.87*	1.16	5.29	−78.07%
Cognitive processes	*9.35*	4.62	7.49	−38.32%
Allure	*7.79*	1.73	5.73	−69.81%
Moralization	*0.20*	0.00	0.00	0%
Summary variables				
Analytic	*44.67*	97.73	96.83	0.93%
Authentic	*60.97*	48.55	68.05	−28.66%

Source(s): Authors' own work

et al., 2017; Park *et al.*, 2010). Similarly, allure-related phrases (e.g. "Buy it before its gone!") were more frequent in experimental ads, supporting their higher persuasive intent (White, 2020; Voorveld and van Noort, 2014).

Moderate differences appeared in Positive Tone and Cognitive Processes, with experimental ads using more emotionally charged and cognitively engaging language than control ads (Boorse *et al.*, 2019). These findings confirm that the experimental manipulation successfully enhanced persuasive elements in CTA imperative messages.

3.5 Ad campaign creation

The campaigns were created within Meta Business ad accounts. A campaign was initiated within the Meta Business account, which included two ad sets (B1 CONTROL, B1 EXPERIMENTAL). One ad set represented the control advertisement, while the other represented the experimental advertisement. Each ad set contained a single advertisement (B1C, B1E), as illustrated in Figure 3.

The campaign was allocated a 14-day run period (from October 10 to October 24, 2023) and a budget of 100 EUR. The campaigns ran simultaneously to minimize temporal effects, such as seasonality (Heien, 2001). The advertisements were continuously active throughout the two weeks. The selected placements were based on the preferred devices of the user groups on their Facebook news feeds (desktop or mobile). The advertisements targeted all operating systems. The Facebook ad parameters are detailed in Table 7.

3.6 Participants

The participants in the experiment were social media users aged between 18 and 60 (Table 8) who spoke Hungarian, resided in Hungary, and met the segmentation criteria associated with the respective advertisements. During the experiment, each user was exposed to a particular advertisement only once. The advertisements reached a total of 13,806 users (n = 13,806) from 35,196 impressions. Impressions count how often a post appears in a timeline, while reach measures unique users who viewed it at least once. Impressions don't indicate engagement.

4. Results

4.1 Descriptive results and metrics

Table 9 presents the descriptive results of the experiment. The descriptive results indicate that the control advertisements performed better in terms of clicks and click-through rate (CTR). The control advertisements had 8.22% more clicks (n = 158 vs. n = 146), and the CTR was 42.47% higher (1.04 vs. 0.73). The relative differences may appear substantial: a 42.47% increase in CTR. However, these descriptive results do not confirm the control ads' superiority but serve as basic campaign insights.

The 14-day campaign concluded with an exportable Facebook A/B split test output, accessible via the "Ads Reports" tab in Ads Manager. Selecting Table 10 metrics ensures an ecologically valid assessment.

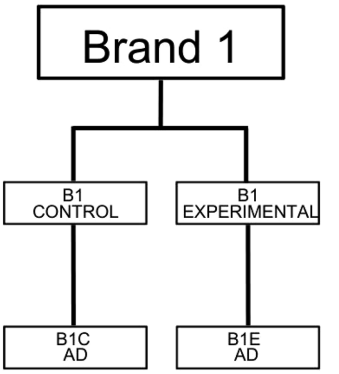


Figure 3. Ad campaign structure. Source: Authors' own work

Table 7. Facebook Ads parameters (C for Campaign, A for Ad set level)

Parameter	Explanation
Buying type (C): auction	Participating in competitive bidding with other advertisers
Campaign objective (C): sales	Facebook's algorithm optimizes for website purchases
Budget optimization (C): none	Budget spend is not optimized
Optimization for ad delivery (A): sales	Facebook's algorithm optimizes for website purchases
When you get charged (A): clicks	The advertiser gets charged for each time a click happens on the ad
Delivery type (A): standard	The ads can show any time of the day

Note(s): The campaign and ad sets had identical parameters
Source(s): Authors' own work

Table 8. Ad reach by age group

Number of reach	Summary
Age groups	
18–24	*1,048*
25–34	*5,127*
35–44	*4,891*
45–54	*2,148*
55–60	*592*

Source(s): Authors' own work

Table 9. Descriptive results of control and experimental ads with relative difference

	Control	Experimental	Relative difference Calculated as (control-experimental)/experimental
Ad impressions The number of times our ads were on screen	15,149	20,047	−24.43%
Reach Number of people seeing the ads	6,452	8,216	−21.47%
Clicks The number of clicks on links within the ad that led to our website	158	146	8.22%
Click-through-rate The percentage of impressions that received a link click out of the total number of impressions	1.04%	0.73%	42.47%

Source(s): Authors' own work

4.2 Data analysis: non-parametric tests and logistic regression on simulated data

CTR offers immediate managerial insight into condition effectiveness. That said, aggregated output data prevent inferential statistical testing. To overcome this limitation, the study suggests two approaches: (1) a non-parametric test, specifically a cross-tabulation comparison using the number of clicks, and (2) a logistic regression analysis on a simulated dataset, where three parameters would be examined. These parameters include reach, where "0" represents the control group and "1" represents the experimental group; clicks, where "0" represents non-

Table 10. Meaning of performance indicators

Performance indicators	Description
Reach:	The number of unique users who were exposed to a given experimental condition. The total reach should be the same across all experimental conditions
Impressions:	The number of times a given experimental condition was displayed. Impressions should be equal to reach to avoid the effects of repetition
Click-Through Rate (CTR):	The ratio of total clicks to total impressions, serving as a quick indicator of whether one condition is more effective than another in driving clicks
Clicks:	The number of clicks associated with different experimental conditions, serving as an indirect measure of user engagement and motivation. This is a key metric used in parameterized testing

Source(s): Authors' own work

clicks and "1" represents clicks; and age, where "0" represents the 18–24 age group, "1" represents the 25–34 age group, "2" represents the 35–44 age group, "3" represents the 45–54 age group, and "4" represents the 55–64 age group. The results of the analysis are presented below.

4.2.1 Chi-square test for clicks. During the experiment, a total of 14,668 reaches were collected. In the cross-tabulation comparison, the proportion of users who clicked or did not click on the advertisements under both experimental conditions was examined. As shown in Table 9, the control condition attracted 158 clicks (and accordingly, 6,294 non-clicks), while the experimental (call-to-action) condition attracted 146 clicks (and accordingly 8,070 non-clicks). Clicks and non-clicks can be specified as columns, with the experimental and control advertisements appearing as rows in a cross-tabulation. This enables a cross-tabulation test (Table 11) to assess outcome dependency on experimental manipulation.

Using the values provided in the previous example, a chi-square test confirmed the difference in proportions, $\chi^2(1) = 8.037$; $p < 0.01$, indicating that the control ad resulted in a higher click-through rate (0.7%) compared to the experimental condition (0.5%). This result suggests that the control condition was more effective in generating user clicks. This supports hypothesis H1, which posits that in social media advertisements where calls to action dominate, users are less likely to click.

4.2.2 Logistic regression on simulated data. The second approach to data analysis involves creating a dataset that simulates clicks and non-clicks among impressions. In the context of a simple A/B test comparing two conditions, this includes: (1) creating a column for the manipulation (control and experimental ads), coded as [0 = control; 1 = experimental], and (2) creating a column for clicks, coded as [0 = No; 1 = Yes]. In the previous example, this means that the first column contains 6,452 instances coded as 0 and 8,216 instances coded as 1. The second column includes 6,294 cases coded as 0 and 158 instances coded as 1 for the

Table 11. Cross-tabulation test

			V2 click 0 NO	1 Yes	Total
V1 Reach	0 Control	Count	6,294	158	6,452
		% within V1 reach	97.6%	2.4%	100.0%
	1 Experimental	Count	8,070	146	8,216
		% within V1 reach	98.2%	1.8%	100.0%
Total		Count	14,364	304	14,668
		% within V1 reach	97.9%	2.1%	100.0%

Source(s): Authors' own work

control condition (where the first column value is 0). For the experimental condition (where the first column value is 1), the second column contains 8,070 instances coded as 0 and 146 instances coded as 1. These data are suitable for running a logistic regression analysis (Table 12), which performs a significance test on the odds ratios. This analysis measures how much more likely participants in the experimental condition are to click compared to those in the control condition.

Similar to the previous findings, the logistic regression analysis yields a positive effect for the control condition regarding clicks. Users exposed to the experimental condition are 0.72 times less likely to click on the advertisement compared to those exposed to the control condition ($B = -0.328$, $SE = 0.12$, $Wald = 7.97$, $p < 0.01$, $Exp(B) = 0.72$).

5. Data analysis and discussion

5.1 Data analysis

Firstly, we applied the chi-square test to compare the proportions of users who clicked on an advertisement versus those who did not. The results confirmed a significant difference in the distribution ($\chi^2(1) = 8.037$, $p < 0.01$). The ratios of clicks to non-clicks were compared for the control condition (Non-clicks = 6,294 vs. Clicks = 158) and the experimental condition (Non-clicks = 8,070 vs. Clicks = 146). The logistic regression analysis conducted on the simulated dataset similarly indicates a positive effect for the control condition regarding the number of clicks, suggesting that users exposed to the experimental condition are 0.72 times less likely to click on the advertisement compared to those exposed to the control condition ($B = -0.328$, $SE = 0.12$, $Wald = 7.97$, $p < 0.01$, $Exp(B) = 0.72$). Both statistical methods support the initial hypothesis (H1) that users not exposed to imperative calls to action (control) were more likely to click on the advertisement. Additionally, users who saw advertisements with imperative calls to action were more likely to exhibit reactance and disregard the ad, reinforcing our initial expectations regarding reactance.

5.2 Discussion

The findings of this study reinforce the narrative introduced earlier (Wang, 2025), where social media users are no longer passive targets but active participants in the battle for their attention. As exposure to these tactics grows, users develop stronger psychological defenses, equipping themselves with invisible shields that not only block persuasive attempts but also send them ricocheting back toward the sender. This backfire effect does more than neutralize the intended influence; in some cases, it leads users to deliberately act against the message's directive, reinforcing reactance rather than compliance. The study demonstrates that CTA imperative messages negatively impact Facebook users' propensity to click on advertisements, with observable reactance during the consumption of Facebook's paid ads. This novel finding contradicts numerous established industry practices (Oltra *et al.*, 2022; Chen *et al.*, 2020; Ren, 2011), as previous research has emphasized the necessity of CTAs and direct imperative messages (Vafainia *et al.*, 2019; Fitzsimons and Lehmann, 2004).

Table 12. Logistic regression analysis

Variables in the equation

	B	S.E.	Wald	df	Sig.	Exp (B)
Reach	−0.328	0.116	7,970	1	0.005	0.721
Constant	−3.685	0.081	2,092,701	1	<0.001	0.025

Source(s): Authors' own work

According to reactance theory, the stronger users perceive the persuasion intent, the more likely they are to resist it (Brehm, 1966). We have expanded Brehm's reactance theory by incorporating the concept of inductive reactance from physics (Maxwell, 1873), which posits that the resistance within a coil, triggered by high voltage (imperative CTA), reduces the effectiveness of persuasion. Consequently, the current lags behind the voltage or may fail to reach it entirely. The mechanism of action is the same in the case of imperative CTAs. The novelty of this concept lies in the creation of the *Inductive Reactance Model (IRM)* (see Figure 1), which posits that the mental inductor is activated by the imperative CTA message, leading to inductive reactance. *Inductive reactance* causes the current in a circuit to lag behind the voltage, and it's exclusively observable in AC circuits. Conversely, *capacitive reactance* leads the current to voltage, and it can also be observed only in AC circuits (see Table 13.). Based on the above, the study claims that by utilizing reverse messages that trigger the discharge of electrical voltage stored in the capacitor (thus activating the capacitor), the generated energy pushes the current forward, making the recipient more likely to be persuaded and thereby driving them towards action. The usage of the reverse message can trigger the phenomena of reverse psychology (Brehm, 1966), which means attempting to achieve the desired response by suggesting the opposite of a specific behavior. This strategy makes use of two psychological concepts – contrast and reactance. The mechanism of reactance was previously presented (Brehm, 1966), and the purpose of contrasting messages was to capture public attention and interest, prompting further investigation (Sinha and Foscht, 2016).

After establishing the theoretical framework to analyze Facebook users' click propensity, Facebook A/B split testing was utilized as an online experimental method, providing a cost-efficient approach for conducting real-world field experiments. The experiment supported the hypothesis that CTA imperative messages in advertisements can trigger reactance and reduce the likelihood of clicking on the ad. This finding is related to the results of Miller *et al.* (2007) and Xu (2014), which describe the negative effects of controlling language, and also connects to the research of Kalny and Walter (2024), who identified the negative perceived perception of imperative messages.

Previous literature (Hu and Wise, 2021; Bilfinger *et al.*, 2023) has primarily focused on ad avoidance and intrusiveness but has not explored the direct link between imperative language and psychological reactance. The IRM fills this research gap, offering a structured explanation of how such language disrupts the persuasive process. The study contributes to the literature by revealing, contrary to many previous studies (e.g. Braca and Dondio, 2023; Oltra *et al.*, 2022; Vafainia *et al.*, 2019), that excessive use of imperative CTA messages can be counterproductive to the advertisement's goal. The findings challenge long-held assumptions, such as those by Fitzsimons and Lehmann (2004), who argued that direct CTAs inherently enhance user confidence and decision-making. The findings enhance the understanding of users' resistance strategies against persuasion and inform advertisers' tactics for neutralizing consumer resistance (e.g. strategies that increase autonomy or reduce intrusiveness).

Table 13. Inductive and capacitive reactance

Types of reactance	Explanation in physics	Reactance tool	Application in communication
Inductive	The currant lags behind the voltage	Inductor (coil)	imperative CTA direct message
Capacitive	The current leads the voltage	Capacitor (able to store electrical charge)	imperative CTA reverse message (reverse psychology)

Source(s): Authors' own work

6. Theoretical and managerial implications

Advertising marksmen carefully aim at social media users, deploying CTA messages to drive engagement. However, to survive on the digital battlefield of persuasion, where invisible psychological shields deflect and even backfire against overly forceful attempts, practitioners must refine their approach. Instead of relying on overtly influential tactics, they must strategically adjust their imperative CTAs, ensuring their messages slip past user defenses rather than triggering reactance. Those who master this balance will continue to thrive in the ever-intensifying competition for attention, while others risk being outmaneuvered in the virtual war of influence.

While similar studies have examined imperatives in advertising, they have often focused on other effects. Dietrich (2018) and El-Sakran and Maklai (2019) studied imperatives in print ads, while Zjakic *et al.* (2017) analyzed their discourse functions without linking them to ad effectiveness. Research by Miller *et al.* (2007) and Xu (2014) explored controlling language, whereas Kalny and Walter (2024) focused on perceived anger and threats to freedom. Labrador *et al.* (2014) and Ren (2011) also examined imperative messages in terms of language. However, no prior research has specifically investigated the effect of CTA imperative messages on reactance, marking a significant research gap that this study addresses.

One of the study's key theoretical contributions is the Inductive Reactance Model, which bridges Psychological Reactance Theory and CTA imperative messages. Applying a physics-based analogy (Maxwell, 1873: inductive and capacitive reactance, see Table 13), the model demonstrates how reactance arises from perceived persuasion pressure. Just as high voltage in an electrical circuit induces resistance, overly forceful CTAs trigger psychological reactance, making users feel their autonomy is threatened (Brehm, 1966). This insight is particularly relevant in the context of increasing social media ad saturation, where advertisers must balance persuasion with user autonomy (Meltwater, 2024). As a result, understanding the consequences of messages in advertisements is becoming an increasingly crucial economic interest for practitioners. Moreover, it provides guidance for researchers, suggesting that the study of advertising messages (potentially through the IRM) could lead to new research findings that enhance our understanding of user perception in social media.

Practically, this study highlights how the classification of imperative messages—such as attention-grabbing, encouraging contact, prompting acquisition, and inviting experience—affects user response (Pennock-Speck and Fuster-Marquez, 2014). Understanding these distinctions helps marketers craft messages that avoid triggering reactance, ensuring their persuasive efforts are more effective.

The IRM also offers novel managerial implications. Reactance can manifest in two forms—inductive or capacitive—each with strategic applications. Capacitive reactance, in particular, aligns with the principles of reverse psychology (Brehm, 1966), suggesting that messages implying the opposite of the intended action may paradoxically achieve better results. Since imperative CTAs often trigger reactance, subtly framed or reverse messages may engage users more effectively, a strategy marketers could leverage to enhance ad performance.

In conclusion, this study provides a novel perspective on the role of CTA imperative messages in digital advertising. The findings suggest that marketers should rethink traditional approaches that favor direct commands and instead adopt strategies that prioritize user autonomy. By doing so, they can reduce resistance, increase engagement, and prevent their messages from backfiring—just as a skilled marksman ensures that every shot reaches its target without unintended recoil.

7. Limitations and future research directions

This study examined Facebook advertisements, which may limit the applicability of findings to other social media platforms with different user behaviors. The sample, consisting solely of Hungarian users, raises questions about generalizability to other cultures. Additionally, using clicks as the primary performance indicator does not capture broader user behaviors, such as long-term engagement, brand loyalty, or shifts in purchase intention.

Future research should explore CTA imperative message effectiveness across different platforms and cultural contexts. Experimental methods like eye-tracking and physiological measures could offer deeper insights into user reactions. Additionally, investigating reverse messages, as suggested by capacitive reactance theory, may offer new directions for advertising strategies. Finally, long-term studies on social media ad effectiveness—including trends in user engagement, brand loyalty, and purchase behavior—would provide a more comprehensive perspective on how imperative CTA messages influence consumers over time.

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